

Mihai Ionita



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[codeberg]

Over 23 years as a software engineer and solution architect, I've built deep expertise across specialized domains - telecommunication and AI - and general-purpose programming, integrating innovative solutions into my work.

Skills

pre-sales delivery system architecture & design team-leading programming teaching
modulith microservices llm ai devops databases containers linux systems administration
webservers application servers load/stress testing

Programming: Java, C, C++, Python, Erlang, Javascript, Spring, Micronaut, GraalVM, TestContainers, Boost, Langchain **Databases:** MySQL, Postgres, H2, sqlite, Cassandra, Oracle, SQL, CQL, GraphQL, OneNDS, Redis **Linux:** Alpine, RedHat, Debian, Wolfi **Documentation:** Markdown, ASCIIDoc, PlantUML, Mermaid **Scripting:** bash, AWK, Perl **DevOps:** Pipelines, Kubernetes, Docker, Terraform, Helm, Ansible, Crossplane, Openstack HOT, K3S, K3D, Skaffold, Vagrant **Pipelines:** dagger.io, Github actions, Azure pipelines, Gitlab CI/CD **Private Cloud:** OpenShift, Openstack, Proxmox, VMWare **Public Cloud:** Azure, AWS, Google, Digital Ocean **Monitoring:** Prometheus, Loki, Grafana, ELK, Micrometer, Open telemetry **Testing:** K6, JMeter, Bruno, Postman/Newman **Security:** Sonar, SpotBugs, Trivy, OWASP, Grype **AI:** Agentic AI workflows, AI Foundry, Azure Cognitive Services, MCP, Hugging Face, Mistral **Protocols:** REST, SOAP, HTTP, TCP, IP, ICMP, Diameter, WebRTC, SIP, Sigtran, SS7, JMX, LDAP, SNMP, SMTP, Syslog **Others:** Chainguard, Nix, ActiveMQ, XML, SAP-CC, JAIN-SLEE, Rhino TAS

Experience

Solution Architect, mindit.io

Mar 2024 - present, Bucharest, Romania

Presales (solution design, estimates) followed by proof-of-concept application development, as well as delivery (high level design, implementation, automated testing and deployment) activities. Technical architecture ownership (end-to-end). I have led teams throughout, driving code reviews, technical direction, and mentoring of team members.

- Helped design and build agentic workflow frameworks for SDLC (software development lifecycle); led teams adopting these frameworks. Used multiple harnesses: Claude Code, GitHub Copilot, OpenCode, Goose, Qwen Code with commands, skills, plugins and MCP servers.
- Created project plan integrations that continuously evolve alongside project

documentation, for both new and existing projects

- Architected a scalable education management system with AI safeguarding features, enabling early detection of issue concerning the education process.
- Designed multi-tenant RAG-based chat-bots for websites, enabling a human-like interaction for the customers' usual queries.
- Incorporated AI-driven solutions, to enhance customer-facing procedures within a B2B agency, ensuring a more responsive initial client experience.
- Created MCP servers for previously un-tackled integrations (both studio (local) and streamable versions).
- Security hardening by: scanning for known vulnerabilities (using Trivy, Gripe, OWASP), enforcing secure deployments with limited attack surface (Chainguard and scratch container images), code review (SonarQube, CodeQL, SpotBugs, etc.), AI-assisted pen-testing.

Technologies used: multi-agent AI, MCP, AI Foundry, Azure Cognitive Services, Azure Container Apps, Terraform, Java, Python, K6, Azure managed Redis, MySQL, bash, powershell

Solution Architect, RSystems

Dec 2020 - Mar 2024, Bucharest, Romania

Presales (solution design, estimates), delivery (from design to implementation and automated testing), support, team-lead and teaching activities.

- Architected custom telecommunication messaging platform, for a tier 1 US operator, designed to maintain synchronization across various systems using multiple retry mechanisms, enabling data-driven decision making to personalize a customer's experience. Designed for micro-services with deployment on Kubernetes. Main drivers: redundancy, speed.
- Tier 3 support activities for containerized production systems using Nokia NCS, spread across multiple clusters
- Solution sizing for cost-effective ramp-up/down, using frameworks that allow a fast container spin-up (e.g. GraalVM + scratch image).
- Redundant deployments for five-nines availability, considering both 3rd party components (including DB fail-over using Cassandra and PostgreSQL) and custom integration software.
- Security hardening by: scanning for known vulnerabilities (using Trivy, Gripe), code review (SonarQube, PMD, SpotBugs).
- Trainer for the company-organized Java Academy (multiple editions).

Technologies used: Java, Flight Recorder, ELK, Kubernetes, NCS, CBIS, Helm, Skaffold, Apache Cassandra, PostgreSQL, JMeter, Postman, bash, Spring, Openshift, Alpine, RedHat, CI/CD pipelines

System Architect, RSystems

Apr 2012 - Dec 2020, Bucharest, Romania

- Designed JAIN-SLEE based call-control solution, on private cloud infrastructure (on-premise), for a Polish telecommunication operator, guiding the team through deployment and ensuring scalability for real-time telecommunication services.
- Led a team to improve a Swiss telecommunication operator's B2B services platform, designing automated charging and reconciliation systems to enhance operational efficiency.
- Led the development of a comprehensive MVNO solution for an operator in the Philippines, ensuring fast time-to-market for a classic telecommunication operator's prepaid plans. The solution facilitated efficient charging (using online charging via SAP-CC convergent charging solution, with Diameter Gy/Ro), call control (using Rhino TAS), and customer service, enabling the marketing department to come up with ever-changing customer prepaid offers.
- Trainer for the company-organized Java Academy (multiple editions). Explained topics ranging from programming to integration to a team of ~10 trainees (each edition).

Technologies used: Java, JAIN-SLEE, Rhino TAS, Erlang, SAP-CC, Openstack, Vagrant, Linux

System Integrator, RSystems

Aug 2011 - Apr 2012, Irving, Texas, US

- Integrated a AAA solution in a tier 1 US telecommunication operator's network.

Technologies used: Nokia OneNDS, OneAAA, LDAP, Diameter, Solaris

System Architect, Computaris - later acquired by RSystems

Dec 2008 - Aug 2011, Bucharest, Romania

- Designed an integration platform between a telecommunication operator and third-party service providers based on a AAA (Authentication, Authorization, Accounting) platform, to create additional revenue streams.
- Created additional components for that platform, interfacing with LDAP and GreenPlum databases.
- Helped design a performance test suite, based on Apache JMeter and led the effort for integrating it.

Technologies used: Java, JMeter, OneNDS, OneAAA, LDAP, GreenPlum, Linux

Senior Software Engineer, Computaris

Mar 2007 - Dec 2008, Bucharest, Romania

- Customer base migration for a telecommunication operator, to a new provisioning platform, without service disruption.
- Provisioning platform development for multi-national operator.

Technologies used: Java, Teligent P90

Software Engineer, Computaris

Mar 2006 - Mar 2007, Bucharest, Romania

- Developed an XML based platform for delivery of new telecommunication services, with very small time-to-market, including charging, provisioning and call-control capabilities.
- Ericsson Diameter charging SDK adaptation, for use in the XML based platform.

Technologies used: C++, Java, Diameter, XML, Linux

Research Engineer, University of the Bundeswehr

Aug 2003 - Mar 2006, Munich, Germany

Delivery (implementation and testing) and support.

- Developed a structural assessment application, using OpenCascade and Sofistik, integrated with wireless communication technology to facilitate real-time measurement input.
- Support activities during application field testing; gathering user feedback and real input data.

Languages

- Romanian: Native
- English: Full professional proficiency
- German: Limited working proficiency
- French: Elementary proficiency

Education

Bachelor in Engineering, Politehnica University of Bucharest

1998 - 2003, Bucharest, Romania

- Information Technology
- CCNA (Cisco Certified Network Administrator) studies

2003, Munich, Germany, [University of the Bundeswehr](#)

- Bachelors dissertation
- Erasmus student exchange program